

Comments

We can participate in HIPE 2027.

For the time being, we should submit this work in some good journal.

- The paper's current form (a shared-task system-description paper) would likely need to be reframed/extended for a journal version.
- The journal selected has to be Hybrid OA model, SCI(E)-indexed, no mandatory APC.
- There are two options:
 - SUBmit the current version with some minor revisions to a conference (but cannot be a shared-task system-description paper as we never participated).

And Submit extended version to a journal

- Submit only one paper to a good journal

Minimum changes required to submit as standalone paper then bound to CLEF: Broader related-work positioning, a fuller comparison against general-purpose RE baselines beyond the shared-task baselines.

1. Information Processing & Management (Elsevier)

2. Expert Systems With Applications (Elsevier)

3. Knowledge-Based Systems (Elsevier)

4. IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)

5. Computer Speech & Language (Elsevier)

1. Aim/Objective & Contributions — 3/5

The methodological novelty is incremental rather than algorithmic. The main contribution is the engineering combination of existing techniques (hmBERT, XLM-R, enriched inputs, class-weighted loss, calibration, and soft-voting ensemble) for the HIPE-2026 shared task. The paper should explicitly clarify what is genuinely novel beyond the integration of known components.

2. Experimental Design — 3/5

Baselines (mBERT, hmBERT, XLM-R) are reasonable and fairly constructed (same dual-head architecture, same loss). However:

- No baseline isolates class-weighted CE vs. unweighted CE — the loss-imbalance design choice itself is never ablated.
- No comparison against focal loss (cited in Sec. 2 as the alternative considered, ref 11), despite explicitly raising it.

- Ablation (Table 4) only varies hmBERT-side components; XLM-R is added as one monolithic ensembling step (A5), so we cannot tell how much of the gain comes from ensembling vs. simply having two models vs. weighting choice ($\alpha=0.60$ is also unablated).

Suggested SOTA/comparable works for benchmarking (beyond the three baselines already used) (LOOK FOR MORE RELEVANT SOTA PAPERS ON THE SELECTED DATASETS. The evaluation is limited to the HIPE-2026 development set. We have to see whether the proposed approach is expected to generalize beyond this benchmark):

1. Schweter et al., hmBERT (already cited, ref 4) — used as encoder but not compared as a fine-tuned-from-scratch RE baseline with a different ensembling strategy.
2. Soares et al., "Matching the Blanks" (entity-marker relation extraction) — directly relevant to the "entity marker tokens" claim in Sec. 2, currently misattributed (see Set A.7).
3. Xu et al., document-level RE with cross-sentence reasoning (already cited, ref 9) — should be empirically compared, not just discussed, since HIPE-2026 is framed as adjacent to document-level RE.
4. A focal-loss variant of the same architecture (ref 11) — the paper claims class-weighted CE was chosen over focal loss based on prior literature, but never empirically validates this for the HIPE-2026 setting.

3. Correctness and Anomalies — 2/5

Found concrete, unresolved arithmetic inconsistencies in Table 3. Since MR is defined as the average of MR_at and MR_isAt, the reported values in Table 3 should satisfy this relationship exactly (up to rounding). Several rows do not, suggesting transcription or reporting errors rather than numerical rounding. These inconsistencies should be corrected before publication. :

- hmBERT (Single): MR = 0.546, MR_at = 0.450, MR_isAt = 0.655 → average = $(0.450+0.655)/2 = \mathbf{0.5525}$, not 0.546.
- hmBERT† (+Calibr.): MR = 0.553, with the *same* MR_at = 0.450 and MR_isAt = 0.655 as the uncalibrated row. By the paper's own metric definition (Eq. 1), MR must equal the average of MR_at and MR_isAt — it cannot change from 0.546 to 0.553 while both component recalls stay fixed. This is mathematically inconsistent.
- XLM-R (Single): MR = 0.538, MR_at = 0.447, MR_isAt = 0.643 → average = $\mathbf{0.545}$, not 0.538. The calibrated row (XLM-R†) then reports MR = 0.545 with the identical MR_at/MR_isAt — i.e., the "0.545" appears to belong to the Single row, not the Calibrated row.
- This pattern strongly suggests transcription errors when populating Table 3 (likely the Single and Calibrated MR values were swapped or miscopied), not just rounding noise.

Additionally, Figure 3 reports MR_isAt = 0.610 for "hmBERT v8 (baseline)," which does not match any value in Table 3 or Table 4 (hmBERT enriched = 0.655 in both). Figure 3's "XLM-R base (fine-tuned)" bar (0.545/0.447/0.643) matches the *calibrated* XLM-R row in Table 3, not the "Single" row (0.538) it's apparently meant to represent. These figure/table mismatches need to be resolved before publication.

4. Generalizability — 4/5

This is already good.

5. Reproducibility — 3/5

The actual *selected* threshold values (τ_p , τ_t , τ_i) from the grid search (Sec. 3.5) are never reported, only the search ranges — this is a meaningful reproducibility gap since calibration is claimed as a measurable contributor (+0.007 MR). No code/data availability statement is present.

Require github repo. The manuscript does not specify:

- random seeds,
- number of training runs,
- hardware used,
- total training time,
- inference time,
- whether the reported numbers are from a single run or averaged.

These details are important for reproducibility.

6. References — 2.5/5

- **Reference 6** (Devlin et al., BERT, NAACL 2019) is cited in Sec. 2 to support the claim that "fine-tuning pre-trained transformers with entity marker tokens has proven highly effective at the sentence level." The original BERT paper does not introduce or specifically validate entity-marker-token relation extraction — this citation is likely misattributed (the standard reference for entity markers in RE is Soares et al., 2019, "Matching the Blanks," which is missing from the bibliography).
- **Reference 11** (Lin et al., Focal Loss, ICCV 2017) is cited to support "prior work (11) has found competitive with focal loss when combined with threshold calibration" — but ref 11 is the original Focal Loss paper, not a comparative study of class-weighted CE vs. focal loss with calibration. This citation appears to support a claim it doesn't actually make.
- **Reference 10** (Inoue, Multi-Sample Dropout, arXiv 2019) appears in the bibliography but is never cited in-body — the multi-sample dropout technique is introduced in Sec. 3.3 without an inline citation, even though the reference exists in the list. This should be cited at "We apply multi-sample dropout ($K = 3$ independent masks)."

Section-Wise Issues

Title Choices. We can look for more

HiPER: An Enriched, Calibrated Transformer Ensemble for Person–Place Relation Extraction from Historical Newspapers

MHIPEX: A Calibrated Transformer Ensemble for Multilingual Person–Place Relation Extraction in Historical Newspapers

Section 3 (Dataset and System Description)

- Eq. 2 and Eq. 4 both reuse the symbol α for two unrelated quantities (pooling interpolation weight = 0.5, and ensemble weight = 0.60). This is a notational clash that could confuse readers comparing equations; recommend renaming one (e.g., β for the ensemble weight in Eq. 4).
- Sec. 3.5 gives threshold *search ranges* but never states the final selected τ values — add a sentence or table reporting the actual calibrated thresholds.

Section 4 (Experimental Setup) The paper doesn't state how the train/dev split interacts with threshold calibration — since thresholds are tuned on the same dev set used for reporting MR, this is a form of dev-set leakage that should at minimum be explicitly named as a limitation here (it is only mentioned, briefly, in the Conclusion).

Section 5.2 (Ablation and Per-Language) A2 and A3 ("Multi-sample dropout" and "CLS+Mean pooling") show *zero* measured MR change (≈ 0), yet the paper still claims them as contributions in the abstract/conclusion ("each component contributes measurably").

According to Table 4, A2 (multi-sample dropout) and A3 (dual pooling) produce little measurable improvement in MR. Require explanation whether these components primarily improve stability rather than average performance and, if so, provide supporting evidence (e.g., variance across runs).

Structure, Flow, and Section Headings

1. **Limitations should be a separate, labeled subsection**, not folded into the final paragraph of the Conclusion. Currently the dev-set leakage caveat and the 3-language caveat are buried as a single paragraph after the summary — pulling these into a "Limitations" subsection (common best practice, e.g., ACL-style papers) would make them more visible to reviewers and readers, and is *more* appropriate than having them appear only at the very end after future work is already discussed.
2. **Section 5.2's per-language results (right half of Table 4) sit awkwardly alongside the ablation study (left half)**. These are two different analyses sharing one table/section. Consider moving the per-language results out of 5.2 and merging them with Section 5.3 (Cross-Dataset and Cross-Lingual Validation), since both address language-level generalization — this would create a cleaner "Generalization" section (in-domain per-language performance + cross-dataset transfer + zero-shot transfer) separate from the pure "Ablation" section.
3. **The threshold calibration mechanism (Sec. 3.5) is described before the experimental results that motivate it (Sec. 5.1, which shows argmax is suboptimal under imbalance)**. Readers encounter the calibration procedure without yet knowing why it's needed; a one-sentence forward-reference in 3.5 to the imbalance statistics already established in Sec. 3.1/Table 1 would tighten this rather than relying on the reader to recall those numbers from two pages earlier.
4. **Section 2 (Related Work) discusses XLM-R's "broader world knowledge" hypothesis** ("we hypothesise complements hmBERT's domain-specific robustness") **before any results exist to support or test it**. This hypothesis is then revisited in Sec.

5.1 ("relation extraction may benefit more from XLM-R's broader world-knowledge representations") and again in Sec. 5.4 (world knowledge as 26% of errors). Consider flagging this explicitly as a hypothesis-to-be-tested in Sec. 2, and then explicitly closing the loop in the Conclusion (currently the Conclusion mentions world-knowledge-driven errors but doesn't tie back to the Sec. 2 hypothesis it originally raised).

Redundancy

1. **Figure 3 and Figure 4 are substantively redundant.** Both show a step-by-step bar chart of MR climbing from a weak baseline to MHIPEX's final ensemble (Fig. 3: mBERT → hmBERT v8 → XLM-R base → hmBERT v12 → MHIPEX; Fig. 4: hmBERT v8 → +DATE/LANG → +multi-sample → +CLS+Mean → +threshold calib → +XLM-R ensemble). They tell overlapping stories at different granularities, and — per Set A.3 — Fig. 3 also contains numbers inconsistent with Table 3. Recommend removing Figure 3 and keeping only Figure 4 (the more granular, table-matching ablation chart), with the mBERT/XLM-R-vs-hmBERT comparison handled entirely by Table 3 in prose.
2. **The per-language performance gap (German low isAt, English low at) is stated three times** with the same underlying numbers: in Sec. 5.2 prose ("German attains the highest at recall (0.486) but the lowest isAt recall (0.599)... English achieves competitive performance (MR = 0.543)"), in the Figure 5 caption, and again in Sec. 5.4 ("consistent with its lower MR_isAt (0.599 vs. 0.672 overall)... consistent with its lower MR_at (0.394 vs. 0.459 overall)"). Suggest stating the per-language numbers once in Sec. 5.2, and in Sec. 5.4 referring back ("as established in Sec. 5.2, German's MR_isAt of 0.599 is well below the 0.672 ensemble average") rather than re-deriving/restating the same comparison.
3. **The hmBERT-outperforms-larger-XLM-R observation is made twice with near-identical framing:** once in Sec. 5.1 ("hmBERT outperforms the larger XLM-R despite having fewer parameters (110M vs. 278M), corroborating the finding of Schweter et al.") and once in Sec. 2 as a forward-looking hypothesis ("hmBERT...demonstrated domain-specific pre-training can outweigh model-scale advantages"). This isn't strictly redundant (hypothesis vs. confirmation) but could be tightened by explicitly cross-referencing rather than re-explaining the Schweter et al. finding in full both times.

Additionally

- The paper reports improvements like +0.007 MR and +0.013 MR. Many reported improvements are relatively small. The paper should report variance across multiple runs or statistical significance to demonstrate that these gains are robust rather than due to random initialization.
- The paper introduces an ensemble. However, nowhere does it discuss the additional computational cost. Since the proposed system requires two transformer models and an

ensemble, the paper should discuss the trade-off between performance gain and computational cost (training/inference time, GPU memory, parameter count).

- Threshold calibration is one of the main contributions. Yet there is almost no qualitative explanation. Since threshold calibration is presented as a major contribution, the paper should include a clearer explanation of why calibration particularly benefits this imbalanced relation extraction task.

Journal

Idea 1: Retrieve structured information for entities from a Knowledge Graph and inject it into the transformer.

Instead of

Sentence
↓
Transformer
↓
Classifier

use

Sentence
+
Entity Linking
+
Knowledge Graph
↓
Knowledge-aware Transformer
↓
Relation Classifier

Candidate KGs

1. Wikidata ★★★★★

Probably the best option.

Useful properties:

- occupation
- country
- capital
- birth place
- death place
- administrative region
- aliases
- historical names

Example:

George Washington

instance of: human
occupation: politician
country: USA
birth place: Virginia

instead of

George Washington

could also be a university
or a street

2. DBpedia

Slightly older but easier to query.

Useful for

- cities
- historical locations
- organizations

3. GeoNames

Especially useful for HIPE.

Contains

- latitude
- longitude
- alternate spellings
- historical names
- administrative hierarchy

Excellent for

Paris

Lyon

Berlin

Vienna

4. Getty Thesaurus of Geographic Names (TGN)

Very good for historical places.

Contains

- historical place names
- alternative spellings
- multilingual names

Perfect for OCR-heavy historical newspapers.

5. World Historical Gazetteer

Very relevant.

Historical

place

→ modern equivalent

→ historical periods

Excellent for historical NLP.

Novel Experiment

Compare

Text only

vs

Text + Wikidata

vs

Text + GeoNames

vs

Text + Historical Gazetteer

Knowledge-Graph-Augmented **at** Classification

Motivation: The authors attribute 26% of manually inspected errors to "implicit world knowledge" — cases where the **at** relation requires facts not stated in the article text (their own example: inferring a politician was "at" the capital from a parliamentary report). This is currently unaddressed; it's only mentioned as future work, never attempted.

Proposed experiment: Before classification, link the person and place mentions to external KG entries (Wikidata is the natural choice — it has structured properties like P19 *place of birth*, P20 *place of death*, P551 *residence*, P39 *position held*, which often encode exactly the person–place associations the **at** relation is trying to recover). Retrieve these facts and inject them into the model either as:

- (a) a concatenated structured-fact string appended to the input (similar in spirit to their existing **<DATE>/<LANG>** token enrichment, so it's a natural methodological extension of what they already do), or
- (b) a separate small encoder/adaptor whose output is fused with the hmBERT/XLM-R representations before the classification head.

This is a clean ablation story: "KG-augmented vs. text-only" on exactly the **probable/true** confusion identified as the dominant error pattern (45.1% probable→true).

Several established resources fit the KG based task well, beyond just Wikidata:

- **Wikidata** — general-purpose KG, best general fit for person↔place facts (birthplace, residence, career posts) as in Experiment 1 above.
- **GeoNames** — a geographic gazetteer/ontology; useful for place disambiguation and normalizing historical place names (helpful since historical place names often differ from modern ones, which the paper doesn't address at all).
- **VIAF (Virtual International Authority File)** — authority records for persons, useful for disambiguating person names, especially the "ambiguous entity names" error category they identify (24% of errors, e.g. "George Washington" person/place collision).
- **Pelagios / Recogito / Peripleo** — a linked-open-data initiative specifically built for historical geography and place-name annotation in digitized texts; closely aligned with the paper's own domain (historical newspapers) and worth citing as directly relevant prior art the paper currently omits.
- **CIDOC-CRM** — a formal ontology for cultural heritage data (events, persons, places, time) used widely in digital humanities; could provide the formal schema for representing the `at/isAt` relations themselves in a more standardized, interoperable way, rather than the paper's current ad hoc three-way/binary label scheme.
- **Europeana Data Model (EDM)** — directly relevant since the paper cites Europeana Newspapers in its own introduction (Sec. 1) as one of the digitization projects motivating this work — but never connects back to Europeana's own metadata/entity infrastructure.

Tying Experiment 1 to Pelagios/GeoNames/Wikidata, and explicitly citing Pelagios as related work. This would also fix a gap in the current version: the current Related Work section (Sec. 2) cites no knowledge-graph or digital-humanities-gazetteer literature at all, despite the paper's own conclusion calling for exactly this kind of integration.

2: OCR-Noise-Aware Contrastive Robustness Training

Motivation: 28% of inspected errors are attributed to OCR corruption (their example: "Philadclphia" for "Philadelphia"). The current pipeline does nothing explicit to address this — it relies entirely on hmBERT's historical pre-training to absorb OCR noise implicitly.

Proposed experiment: Synthetically inject realistic OCR-style character confusions (common substitutions like `rn↔m`, `l↔1`, `c↔e`, `long-s↔f`, digit/letter swaps) into training data, then train with a contrastive or consistency objective that pulls clean and OCR-corrupted versions of the same entity mention together in representation space. Ablate against the current model to see whether this measurably improves recall specifically on the OCR-corruption-flagged subset of their own 50-example error analysis (which they could re-annotate as a small diagnostic test set — itself a methodological contribution, since right now that 50-example analysis is descriptive only, not used as an evaluation set).

The paper itself identifies that many errors require **external world knowledge**:

- George Washington (person vs place)
- politicians and capitals
- historical place names
- OCR-induced ambiguity

Currently the model only uses text.

3: Ontology-Constrained Relation Prediction

Instead of predicting relations independently, use ontology rules.

Example:

Suppose

Person

isAt

City

is possible.

But

Person

isAt

River

is impossible.

The ontology can eliminate impossible predictions.

Example

Suppose model predicts

Einstein

isAt

Atlantic Ocean

Ontology says

Human

cannot be located at

Ocean

Prediction rejected.

Or

French Parliament

isAt

Paris

valid.

Ontology sources

CIDOC CRM ★★★★★

Very famous cultural heritage ontology.

Widely used for museums and historical archives.

Excellent fit for historical documents.

Europeana Data Model

Historical archive ontology.

Linked Open Data cloud

Can connect

person

city

organization

country

Experiment

Compare

Transformer

vs

Transformer + ontology validation

vs

Transformer + ontology inference

This is genuinely novel.

4. Another Idea

The paper currently trains relation extraction only.

Instead, jointly learn

NER

+

Relation Extraction

+

Entity Linking

in one model.

Document

↓

NER

↓

Entity Linking

↓

Relation Extraction

The linked entity provides knowledge.

This has become popular recently and would be much more innovative than simply ensembling two transformers.

5. Retrieval-Augmented Relation Extraction (RAG)

When encountering

Napoleon

retrieve

Napoleon

French emperor

France

Paris

from Wikidata.

Append retrieved facts.

Transformer predicts relation.

This is becoming popular but is still relatively unexplored for historical newspapers.

6. Graph Neural Network over Entities

Instead of treating entity pairs independently,

construct

Document

↓

Entity Graph

↓

Graph Attention Network

↓

Relation Prediction

Nodes

person

city

country

organization

Edges

co-occurrence

sentence proximity

dependency links

document links

This captures global document context.

Which Knowledge Sources Best Fit This Paper?

Resource	Suitable	Why
Wikidata	★★★★★	Rich semantic facts, multilingual, easy entity linking
GeoNames	★★★★★	Geographical hierarchy and alternate place names
Getty TGN	★★★★★	Historical place names and multilingual variants
World Historical Gazetteer	★★★★★	Designed for historical geographic entities
DBpedia	★★★★	General-purpose structured knowledge
CIDOC CRM	★★★★	Ontology for cultural heritage reasoning
Europeana EDM	★★★★	Historical archive semantics